PARISHRAM 2027

Electric Charges and Fields

LECTURE-1 PHYSICS

Rakshak sir

Topics To Be Covered

1. ... Introduction To Class XII Physics
2. ... Electric Charges and its Properties
3. ...
4. ...

Rakshak Sir
JOIN MY OFFICIAL TELEGRAM CHANNEL
@RAKSHAKSIRSCIENCETG

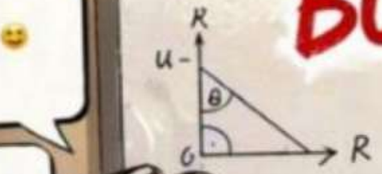
NOT JUST A TEACHER, BUT A MENTOR FOR LIFE!

@aditya.yadav_18 ⭐⭐⭐⭐⭐
Sir ki wajah se Physics
ab darawani nahi lagti! ❤️
Concept itna easy bana dete ho 😊
Thank you so much Sir!

@anushkaverma_10 ⭐⭐⭐⭐
Boards mein 95/100 in Physics! 🔥
Sir aapki strategy, notes aur
teaching style sab best hai!
Aapne baccha liya 🙌

@manasijoshi_ ⭐⭐⭐⭐⭐
Rakshak Sir ke one shots =
Game Changer 🔥
Itna crisp, itna clear explanation
kahin nahi milta!
Last day revision se full marks aa gaye 😭❤️

@pratham_120 ⭐⭐⭐⭐⭐
Sir sirf padhate nahi,
hum students ki problems samajhte hain.
Best mentor ever! 😭❤️



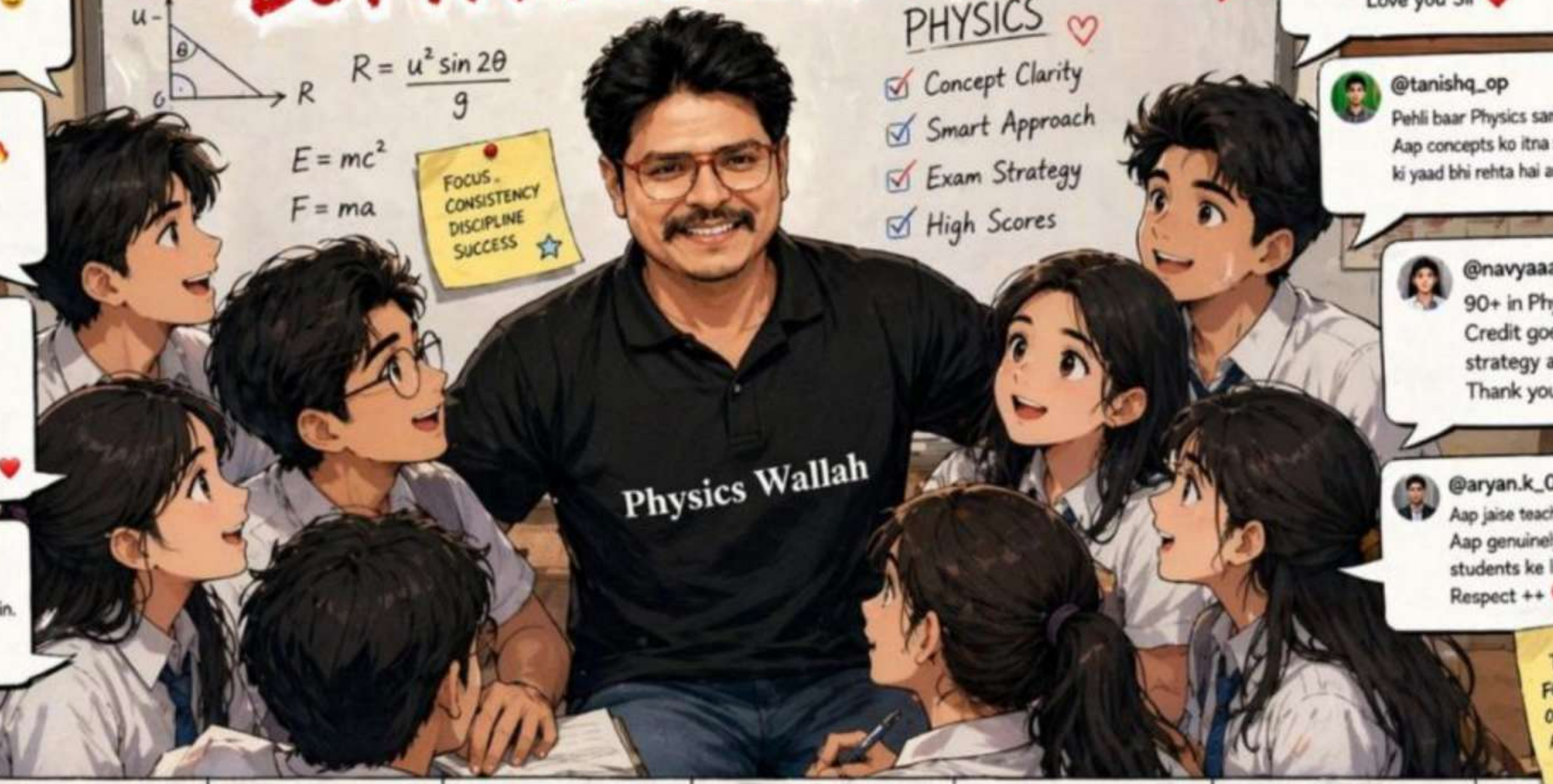
$$R = \frac{u^2 \sin 2\theta}{g}$$

$$E = mc^2$$

$$F = ma$$

FOCUS.
CONSISTENCY
DISCIPLINE
SUCCESS

PHYSICS
- ☑ Concept Clarity
- ☑ Smart Approach
- ☑ Exam Strategy
- ☑ High Scores



@sakshisingh_ ⭐⭐⭐⭐⭐
Sir ki class ekdum alag level ki hoti hai 🔥
Energy, explanation, examples
sab kuch perfect! 🙌
Love you Sir ❤️

@tanishq_op ⭐⭐⭐⭐
Pehli baar Physics samajh aayi Sir ki wajah se!
Aap concepts ko itna simple bana dete ho
ki yaad bhi rehta hai aur maza bhi aata hai 😊❤️

@navyaaaaa_ ⭐⭐⭐⭐⭐
90+ in Physics in Boards 😍
Credit goes to Rakshak Sir's
strategy and motivation!
Thank you Sir 🙏❤️

@aryan.k_07 ⭐⭐⭐⭐⭐
Aap jaise teachers bahut kam hote hain Sir!
Aap genuinely care karte ho
students ke liye.
Respect ++ ❤️

THANK YOU SIR
FOR MAKING PHYSICS
OUR FAVOURITE SUBJECT
AND BELIEVING IN US!

Our Journey Throughout the Year

April	May - June	July - August	September - October	November	December - January
New Beginnings	Concept Building	Practice & Doubts	Revision & Tests	Pre-Board Preparation	Board Exams
Basics clear kiye, foundation strong ki.	Har concept ko deep samjha.	Problems solve kiye, doubts clear kiye.	Regular tests aur revision se confidence badhaya.	Weak areas cover kiye, strategy sharp ki.	Mehnat ka result, dreams achieved! 🎉

BECAUSE OF YOU, WE DID IT! ❤️

THANK YOU RAKSHAK SIR FOR YOUR GUIDANCE, SUPPORT AND UNCONDITIONAL BELIEF IN US. YOU ARE THE REASON WE AIM HIGH! 🙌❤️

Class XII Ki Shuruaat

Q1) Class XI Barbaad ? ⟹ Go full for class 12th
⟹ Bridge Course (Basic Maths)
⟹ Jo use hogo, wohi Padha Denge

Q2) Class XII Boards Tough ?
→ Class XII tough ✗
→ Class XII Boards tough ✓

Q3) Plan to Score
→ Manifest $\frac{100}{100}$

My promise :-

1) Start from Basics [Y.K.B.]

2) Tricks for CUET/NDA (RDx) NEET/JEE

3) Practice Questions

After Syllabus Completion

"Practice Sessions"
CUET spl.
NDA spl.

CBQ
AR

Practice Session after each Chapter (PYQs)

4) I will offer My full dedication and discipline in this year's Parishram Batch

Signature

Your Promise :-

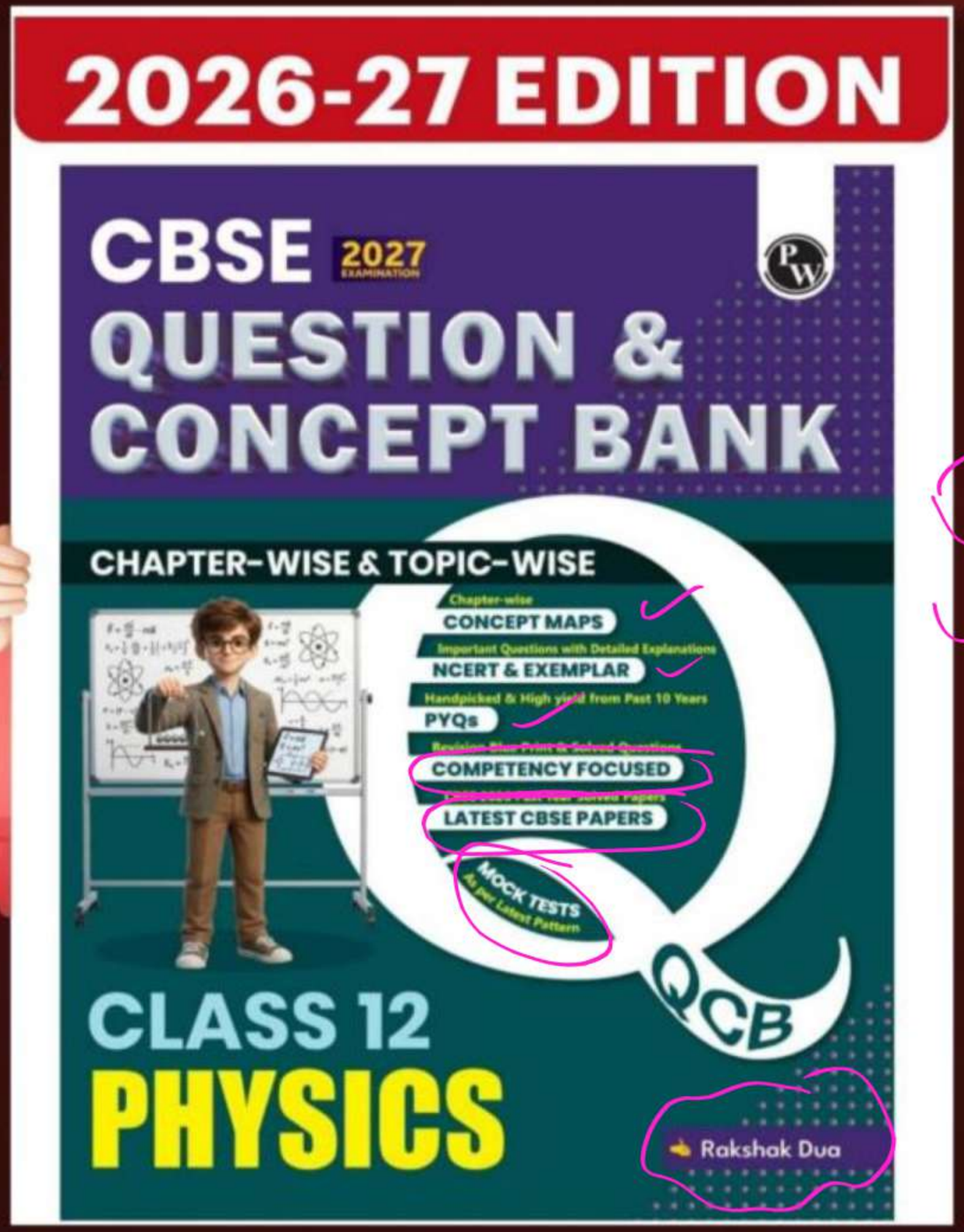
1) Do not Miss any class

2) Try & understand the Concepts, Short Notes for Revision

3) Roz Ka Kaam Roz

4) I will offer "My full dedication and discipline this Year

Signature



Amazon
Flipkart
PW store

70

Class XII (2025-2026)

Physics (Theory)

Time: 3 hrs.

Max Marks: 70

		Marks
Unit-I	**Electrostatics**	
	Chapter-1: Electric Charge and fields	5
	Chapter-2: Electrostatic Potential and Capacitance	5
Unit-II	**Current Electricity**	
	Chapter-3: Current Electricity	6

10 Marks

Electric Charges and Fields

TOPICS : "5 Marks"

Electric charges. Conservation of charge, Coulomb's law-force between two-point charges, forces between multiple charges; superposition principle and continuous charge distribution. Electric field, electric field due to a point charge, electric field lines, electric dipole, electric field due to a dipole, torque on a dipole in uniform electric field.

YKB

Electric flux, statement of Gauss's theorem and its applications to find field due to infinitely long straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell (field inside and outside).

Electric Charge

entity ×

property ✓

← ⊕ R ⊕ →

← ⊖ R ⊖ →

⊕ A ⊖

→ ←

Its is an intrinsic property of the elementary particles like electrons, protons, etc. of which all the objects are made up of.

→ Algebraic Sum

Its is a scalar quantity.

It SI unit is Coulomb (C).

$e = 1.6 \times 10^{-9}$ coulomb

$1mC = 10^{-3}C$

$1\mu C = 10^{-6}C$

$1nC = 10^{-9}C$

$1pC = 10^{-12}C$

$Q_{net} = +10-12$

$= -2C$

Net charge

+10C

−12C

(equal + & − charges)

No ⇐ Chargeless ?

YES ⇐ Neutral ?

Photon

→ Mass = 0

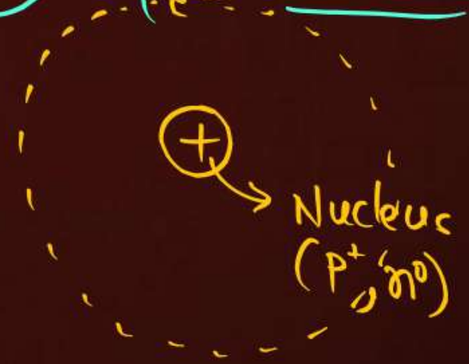
→ Charge = 0

Y.K.B.

$1e^- = -1.6 \times 10^{-19}C$ ✓

$1p^+ = +1.6 \times 10^{-19}C$ ✓

$1n$ = Zero Coulomb



HOMEWORK

Promise Poora Karna!

Thank You